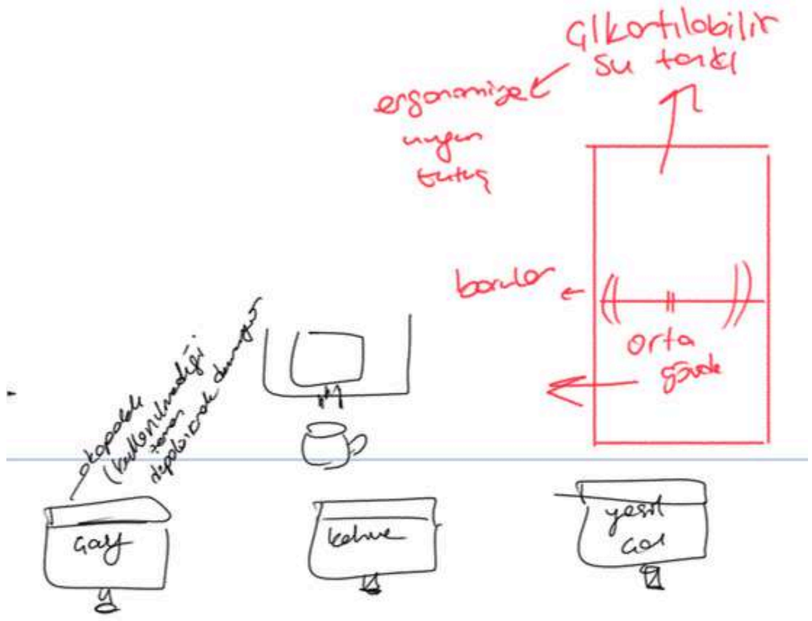
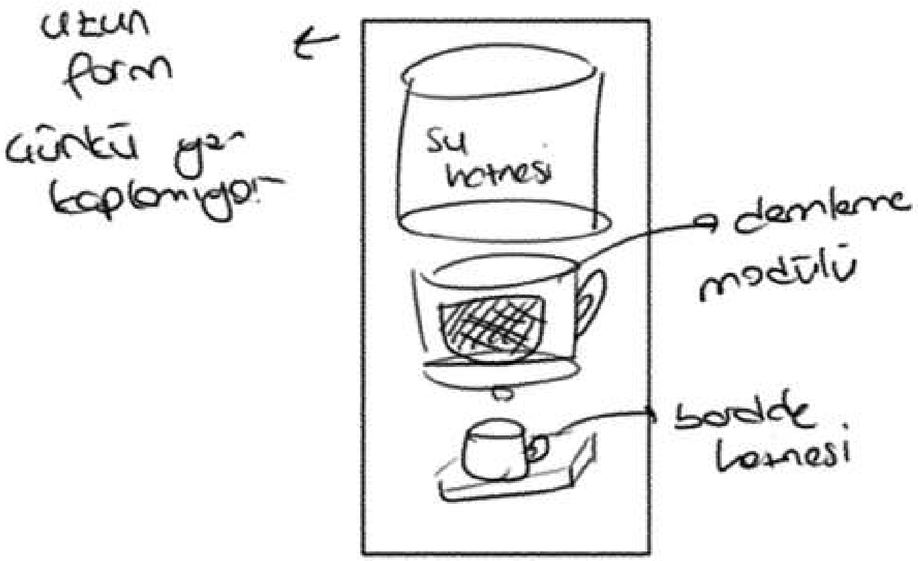
At this stage, I started to think about the general structure of the machine.
I wanted the product to have a vertical form so it would take less space on the counter.
I placed the water tank at the top, the brewing module in the middle, and the cup area at the bottom.

This idea helped me understand the basic layout of the machine.
However, I realized that this structure might become too tall, but I decided to continue with this direction because the vertical form fits small living spaces.





Decision:

Accepted → vertical layout
Accepted → top water tank
Concern → too tall form

AI Role:

Re-framer

Keywords:
vertical form, compact design, layout, structure

After defining the idea of a compact brewing machine, I used AI to generate the first form studies.
At this stage, I wanted to understand how a vertical structure could work.
The result looked clean, but the system was too simple and did not support modular brewing.

Decision:

Rejected → too simple
Accepted → vertical form

AI Role: Expander

Then I focused on the internal system of the machine.
I needed the water to move from the tank to the brewing module.

With AI support, I designed the internal tube system, heating unit, and diffuser.

This stage made the design technically believable.

Decision
Accepted → internal flow
Accepted → heating unit
Accepted → diffuser

AI Role → Clarifier

After solving the internal system, I realized that tea and coffee need different brewing methods.

Because of this, I designed interchangeable brewing modules.

Each module uses a different brewing structure depending on the drink.

Decision
Accepted → modular system
Rejected → single module

AI Role → Re-framer

In this stage, I designed the internal parts of the module.
Coffee needs a shower head and filter paper.
Tea needs a mesh diffuser.

Because of this, I separated the insert parts.

Decision
Accepted → two diffusers
Rejected → same insert

AI Role → Critique

Later, I noticed that the machine was too tall.
I redesigned the proportions and made the body more compact.

I also added control buttons and hot water output.

Decision
Accepted → compact form
Accepted → control panel
Rejected → tall form

AI Role → Critique

At this stage, I focused on the hot water outlet area.

I wanted the user to be able to use hot water without using the brewing module.
Because of this, I designed a separate hot water spout.

Instead of placing the cup outside the machine, I created a recessed space inside the body.
This allowed the cup to be placed under the hot water outlet without increasing the overall size of the product.

This solution helped keep the machine compact while adding an extra function.

Keywords
hot water outlet
space saving
extra function
compact body

Decision
Accepted → separate hot water spout
Accepted → inner space for cup
Rejected → external extension

AI Role
Critique

After defining the structure, I focused on usability.

The middle brewing module needed to be removable for cleaning and for changing between tea and coffee.

To make this easier, I added a handle to the module.
The handle allows the user to hold the module comfortably while attaching or removing it.

I also simplified the internal parts so the module can be cleaned easily.

Keywords
ergonomics
handle
removable module
cleaning
usability

Decision
Accepted → handle
Accepted → removable module
Accepted → easy cleaning
Rejected → fixed module

In the final stage, I defined the technical dimensions of the product.

The goal was to keep the machine compact while still fitting the water tank, heating unit, and brewing module.

The final dimensions allow the product to be used comfortably on a kitchen counter without taking too much space.

Height: 420 mm
Width: 180 mm
Depth: 300 mm

These proportions keep the product stable, compact, and suitable for small kitchens.

Reflection:

Defining the dimensions helped balance usability, stability, and internal space.

Keywords
dimensions
proportion
technical design
compact

AI Role
Clarifier

Throughout the project, I used AI to explore forms, solve technical problems, and visualize ideas.

However, the final decisions were based on usability, size, and realism.

The most difficult parts were:

designing the modular system
creating a compact form
solving the water flow
adding extra functions without increasing size

This process showed that AI is useful for exploration, but the designer must control the final result.

AI Roles
Clarifier
Expander
Critique
Re-framer